

Time & Causality

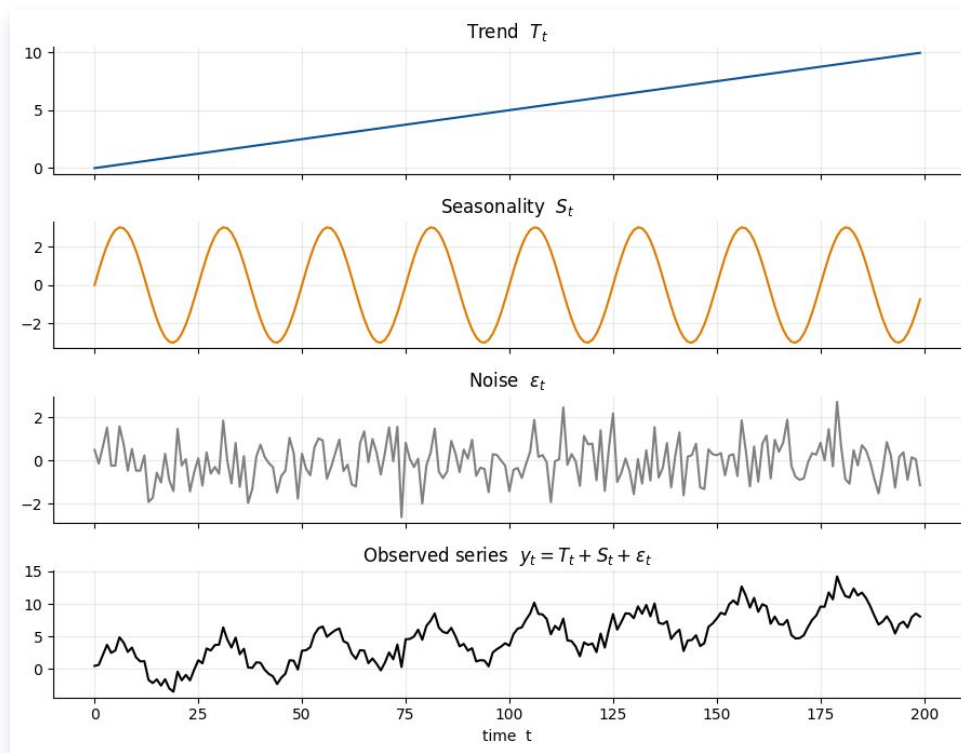
Unit 5

Time Series

A time series is a sequential set of data points, measured typically over successive times.

Time series analysis comprises methods for analyzing time series data in order to extract meaningful statistics and other characteristics of the data.

A sequence of observations recorded over time: $y_1, y_2, y_3, \dots, y_n$



Categories of Time Series

- **univariate vs. multivariate**

A time series containing records of a single variable is termed as univariate, but if records of more than one variable are considered then it is termed as multivariate.

- linear vs. non-linear

A time series model is said to be linear or non-linear depending on whether the current value of the series is a linear or non-linear function of past observations.

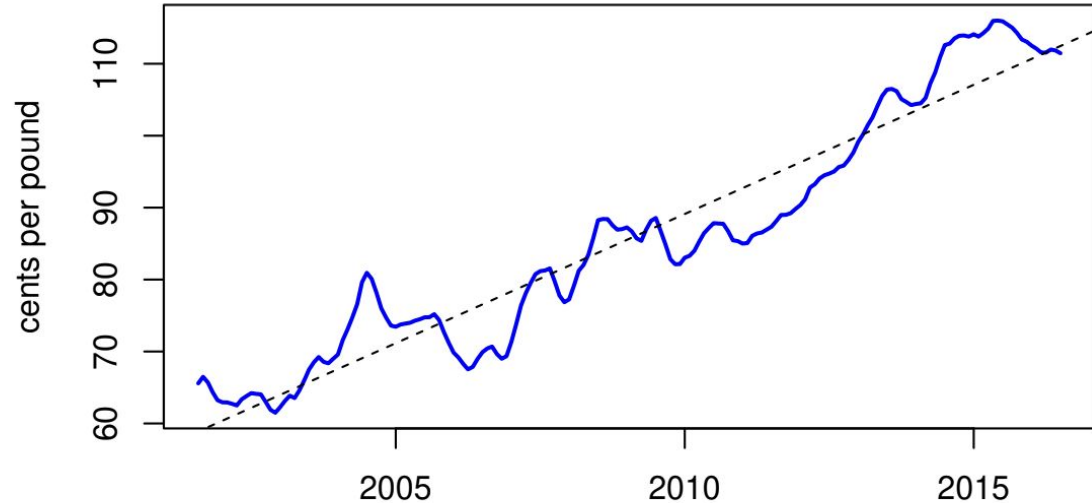
- discrete vs. continuous

In a continuous time series observations are measured at every instance of time, whereas a discrete time series contains observations measured at discrete points in time.

Components of Time Series

Trend

The general tendency of a time series to increase, decrease or stagnate over a long period of time.

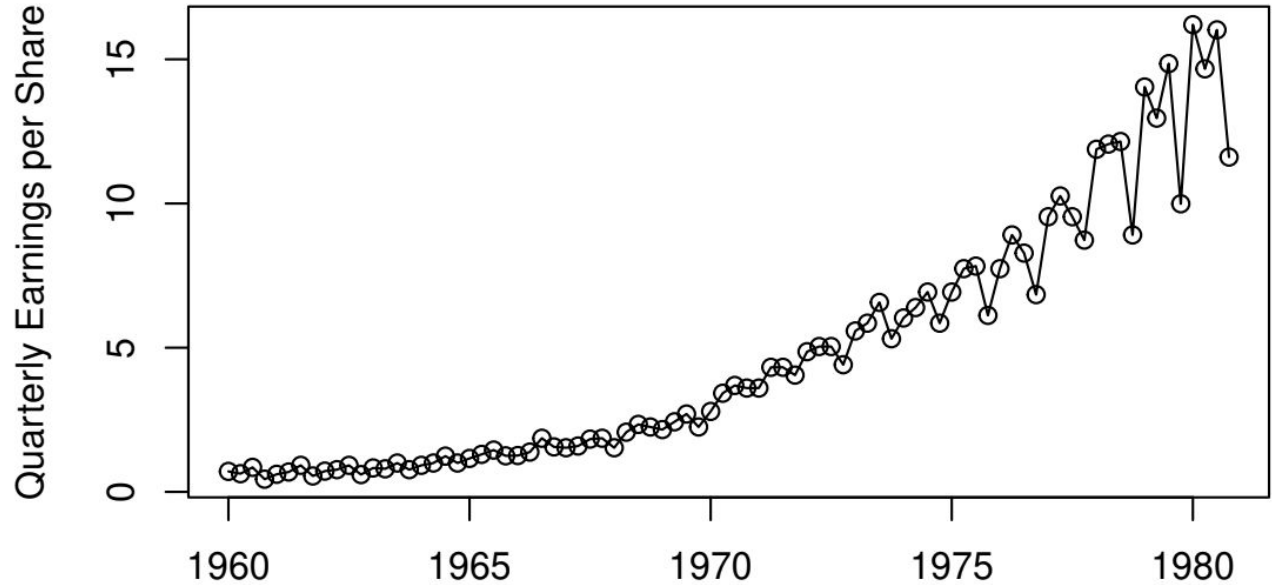


The price of chicken: monthly whole bird spot price, Georgia docks, US cents per pound, August 2001 to July 2016, with fitted linear trend line.

Components of Time Series

Seasonal variation

This component explains fluctuations within a year during the season, usually caused by climate and weather conditions, customs, traditional habits, etc.

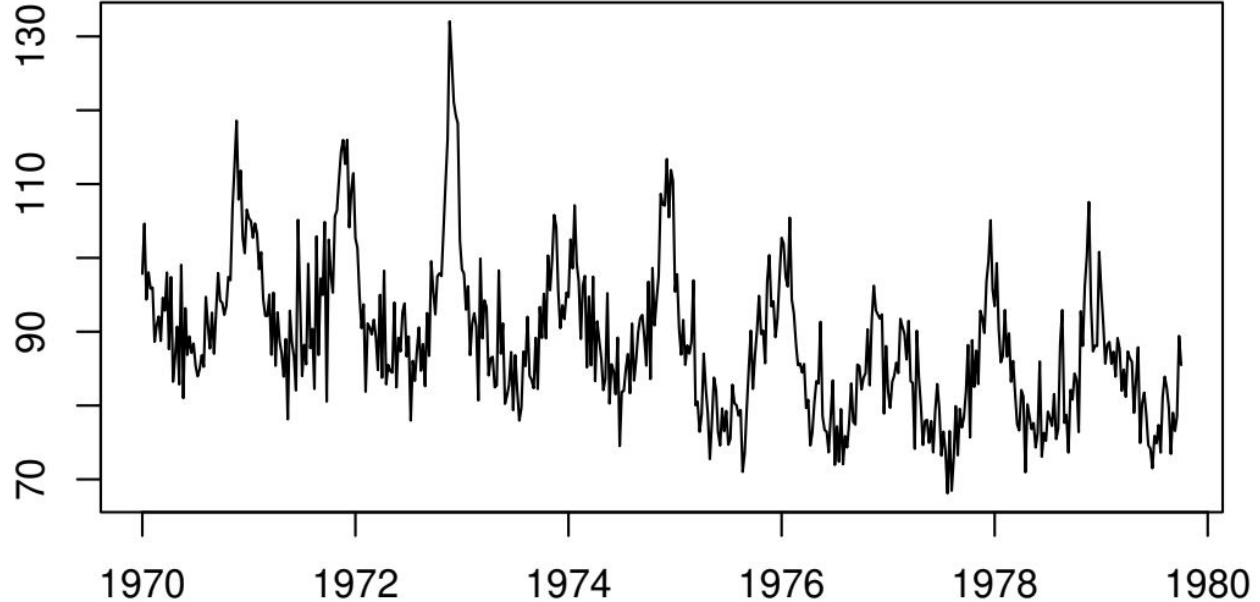


Components of Time Series

Cyclical variation

This component describes the medium-term changes caused by circumstances, which repeat in cycles.

The duration of a cycle extends over longer period of time.



Components of Time Series

Irregular variation

Irregular or random variations in a time series are caused by unpredictable influences, which are not regular and also do not repeat in a particular pattern.

These variations are caused by incidences such as war, strike, earthquake, flood, revolution, etc.

There is no defined statistical technique for measuring random fluctuations in a time series.

Combination of four components

- Additive Model

$$Y(t) = T(t) + S(t) + C(t) + I(t)$$

Assumption: These four components are independent of each other.

- Multiplicative Model

$$Y(t) = T(t) \times S(t) \times C(t) \times I(t)$$

Assumption: These four components of a time series are not necessarily independent and they can affect one another.

- **White Noise**

- A simple time series could be a collection of **uncorrelated** random variables, $\{w_t\}$, with zero mean $\mu = 0$ and finite variance σ_w^2 , denoted as $w_t \sim wn(0, \sigma_w^2)$.

- **Gaussian White Noise**

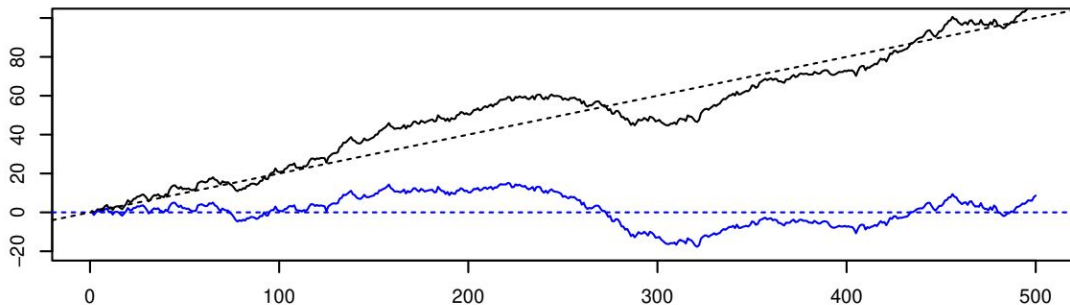
- A particular useful white noise is Gaussian white noise, wherein the w_t are **independent normal random variables** (with mean 0 and variance σ_w^2), denoted as $w_t \sim \text{iid } \mathcal{N}(0, \sigma_w^2)$.

- White noise time series is of great interest because if the stochastic behavior of all time series could be explained in terms of the white noise model, then classical statistical methods would suffice.

Time Series Example: Random Walk

- A random walk is the process by which randomly-moving objects wander away from where they started.
- Consider a simple 1-D process:
 - The value of the time series at time t is the value of the series at time $t - 1$ plus a completely random movement determined by w_t . More generally, a constant drift factor δ is introduced.

$$X_t = \delta + X_{t-1} + w_t = \delta t + \sum_{i=1}^t w_i$$



Measurement Functions

- **Mean function**

- The mean function is defined as

$$\mu_t = \mu_{X_t} = E[X_t] = \int_{-\infty}^{\infty} xf_t(x)dx,$$

provided it exists, where E denotes the usual expected value operator.

- Clearly for white noise series, $\mu_{w_t} = E[w_t] = 0$ for all t .
- For random walk with drift ($\delta \neq 0$),

$$\mu_{X_t} = E[X_t] = \delta t + \sum_{i=1}^t E[w_i] = \delta t$$

Auto Covariance for Time Series

- Lack of independence between adjacent values in time series X_s and X_t can be numerically assessed.
- **Autocovariance Function**
 - Assuming the variance of X_t is finite, the autocovariance function is defined as the second moment product

$$\gamma(s, t) = \gamma_X(s, t) = \text{cov}(X_s, X_t) = E[(X_s - \mu_s)(X_t - \mu_t)],$$

for all s and t .

- Note that $\gamma(s, t) = \gamma(t, s)$ for all time points s and t .
- The autocovariance measures the **linear dependence** between two points on the same series observed at different times.
 - Very smooth series exhibit autocovariance functions that stay large even when the t and s are far apart, whereas choppy series tend to have autocovariance functions that are nearly zero for large separations.

Auto Correlation for Time Series

- **Autocorrelation Function (ACF)**

- The autocorrelation function is defined as

$$\rho(s, t) = \frac{\gamma(s, t)}{\sqrt{\gamma(s, s)\gamma(t, t)}}$$

- According to Cauchy-Schwarz inequality

$$|\gamma(s, t)|^2 \leq \gamma(s, s)\gamma(t, t),$$

it's easy to show that $-1 \leq \rho(s, t) \leq 1$.

- ACF measures the **linear predictability** of X_t using only X_s .
 - If we can predict X_t perfectly from X_s through a linear relationship, then ACF will be either $+1$ or -1 .

Stationarity for Time Series

- Forecasting is difficult as time series is non-deterministic in nature, i.e. we cannot predict with certainty what will occur in the future.
- But the problem could be a little bit easier if the time series is **stationary**: you simply predict its statistical properties will be the same in the future as they have been in the past!
 - A stationary time series is one whose statistical properties such as mean, variance, autocorrelation, etc. are all **constant over time**.
- Most statistical forecasting methods are based on the assumption that the time series can be rendered approximately stationary after mathematical transformations.

Auto correlation for time series

- Recall that the autocovariance $\gamma_X(s, t)$ of stationary time series depends on s and t only through $|s - t|$, thus we can rewrite notation $s = t + h$, where h represents the time shift.

$$\gamma_X(t + h, t) = \text{cov}(X_{t+h}, X_t) = \text{cov}(X_h, X_0) = \gamma(h, 0) = \gamma(h)$$

- Autocovariance Function of Stationary Time Series**

$$\gamma(h) = \text{cov}(X_{t+h}, X_t) = E[(X_{t+h} - \mu)(X_t - \mu)]$$

- Autocorrelation Function of Stationary Time Series**

$$\rho(h) = \frac{\gamma(t + h, t)}{\sqrt{\gamma(t + h, t + h)\gamma(t, t)}} = \frac{\gamma(h)}{\gamma(0)}$$

Partial Auto Correlation

- Another important measure is called partial autocorrelation, which is the correlation between X_s and X_t with the linear effect of “everything in the middle” removed.
- **Partial Autocorrelation Function (PACF)**
 - For a stationary process X_t , the PACF (denoted as ϕ_{hh}), for $h = 1, 2, \dots$ is defined as

$$\phi_{11} = \text{corr}(X_{t+1}, X_t) = \rho_1$$

$$\phi_{hh} = \text{corr}(X_{t+h} - \hat{X}_{t+h}, X_t - \hat{X}_t), \quad h \geq 2$$

where $\hat{X}_{t+h}$ and $\hat{X}_t$ is defined as:

$$\hat{X}_{t+h} = \beta_1 X_{t+h-1} + \beta_2 X_{t+h-2} + \dots + \beta_{h-1} X_{t+1}$$

$$\hat{X}_t = \beta_1 X_{t+1} + \beta_2 X_{t+2} + \dots + \beta_{h-1} X_{t+h-1}$$

- If X_t is Gaussian, then ϕ_{hh} is actually conditional correlation

$$\phi_{hh} = \text{corr}(X_t, X_{t+h} | X_{t+1}, X_{t+2}, \dots, X_{t+h-1})$$

ARIMA

ARIMA is an acronym that stands for Autoregressive Integrated Moving Average.

- AR Autoregression. A model that uses the dependent relationship between an observation and some number of lagged observations.
- I Integrated. The use of differencing of raw observations in order to make the time series stationary.
- MA Moving Average. A model that uses the dependency between an observation and a residual error from a moving average model applied to lagged observations.
- Each of these components are explicitly specified in the model as a parameter.
- Note that AR and MA are two widely used linear models that work on stationary time series, and I is a preprocessing procedure to stationarize time series if needed.

ARIMA Notations

- A standard notation is used of $ARIMA(p, d, q)$ where the parameters are substituted with integer values to quickly indicate the specific ARIMA model being used.
 - p The number of lag observations included in the model, also called the lag order.
 - d The number of times that the raw observations are differenced, also called the degree of differencing.
 - q The size of the moving average window, also called the order of moving average.
- A value of 0 can be used for a parameter, which indicates to not use that element of the model.
- In other words, ARIMA model can be configured to perform the function of an ARMA model, and even a simple AR, I, or MA model.

ARMA and ARIMA models

AR(p) — Autoregressive

today = weighted sum of its own past values + noise

$$y_t = c + \phi_1 y_{t-1} + \dots + \phi_p y_{t-p} + \varepsilon_t$$

MA(q) — Moving Average

today = weighted sum of past shocks (errors)

$$y_t = \mu + \varepsilon_t + \theta_1 \varepsilon_{t-1} + \dots + \theta_q \varepsilon_{t-q}$$

ARMA(p, q) — both combined

$$y_t = c + \sum_{i=1}^p \phi_i y_{t-i} + \varepsilon_t + \sum_{j=1}^q \theta_j \varepsilon_{t-j}$$

ARIMA(p, d, q) — difference d times, then ARMA

$$(1 - L)^d y_t \sim \text{ARMA}(p, q)$$

Reading (p, d, q)

AR

p

memory of past values

I

d

differences to reach stationarity

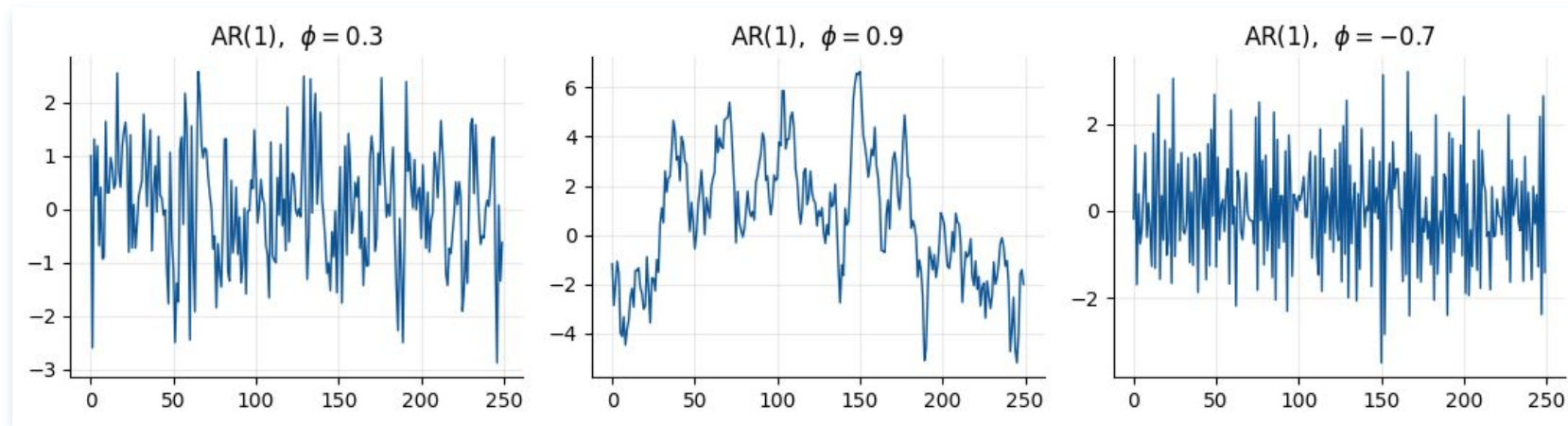
MA

q

memory of past errors

Lag operator: $Ly_t = y_{t-1}$

AR(1) behaviour: $y_t = \phi y_{t-1} + \varepsilon_t$



$$\phi \approx 1$$

Long memory — the series wanders slowly and takes a long time to return.

$$\phi \approx 0$$

Almost no memory — looks essentially like white noise.

$$\phi < 0$$

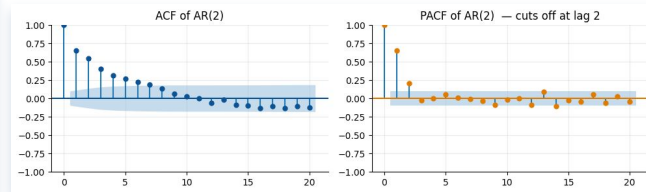
Rapid up-and-down oscillation from one step to the next.

Reading ACF & PACF: the Box–Jenkins rule

Model	ACF	PACF
AR(p)	tails off gradually	cuts off after lag p
MA(q)	cuts off after lag q	tails off gradually
ARMA(p,q)	tails off	tails off

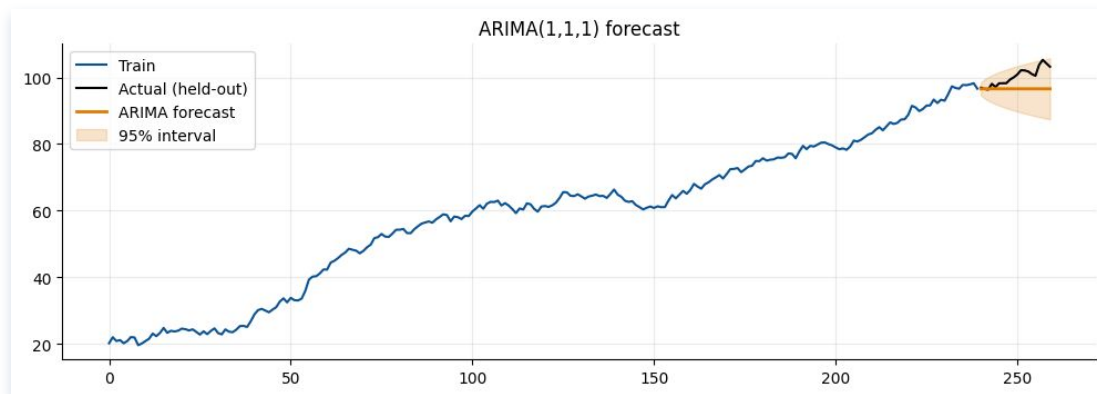
Key idea

A sharp cut-off tells you the order. In the AR(2) example on the right, the PACF drops to near zero after lag 2 — so we read $p = 2$. The ACF tails off smoothly, as expected for an AR process.



AR(2): PACF cuts off at lag 2

ARIMA in action: fit & forecast



ARIMA(1,1,1) forecast with 95% interval on a held-out tail

1

Identify

Check stationarity (ADF); read ACF/PACF for candidate p , q ; set d by differencing.

2

Estimate

Fit the $ARIMA(p, d, q)$ coefficients by maximum likelihood.

3

Diagnose

Residuals should look like white noise (no leftover autocorrelation).

4

Forecast

Project forward with prediction intervals.

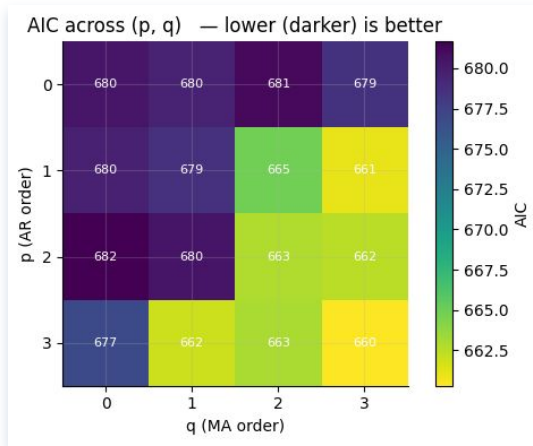
Choosing the model: AIC & BIC

Balance goodness-of-fit against complexity. Lower is better.

$$\text{AIC} = 2k - 2\ln(\hat{L})$$

$$\text{BIC} = k\ln(n) - 2\ln(\hat{L})$$

- $-2 \ln L$ rewards fit — a higher likelihood lowers the score.
- $2k$ (AIC) or $k \ln n$ (BIC) penalises the number of parameters k .
- Since $\ln n > 2$ once $n \geq 8$, BIC penalises complexity more heavily — it tends to pick simpler models.
- Grid-search over (p, q) , then choose the lowest-scoring model.



AIC across (p, q) — darker = better

5.2

Causal Analysis

From “what will happen?” to “what happens if we intervene?”

Causation vs. correlation · Granger causality · Causal DAGs



5.2.1

Correlation $\neq$ Causation

$X \rightarrow Y$

Real causation

$Y \rightarrow X$

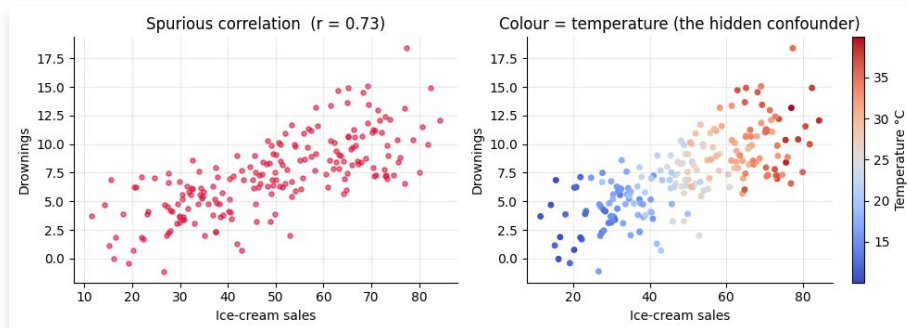
Reverse causation

$Z \rightarrow X, Z \rightarrow Y$

A confounder drives both

coincidence

Pure chance



The classic trap

Ice-cream sales and drownings are strongly correlated — but neither causes the other. Hot weather (temperature) drives both. In the code they share no direct link, yet r is large.

Granger causality

X Granger-causes Y if the past of X helps predict Y beyond Y's own past. Compare two models:

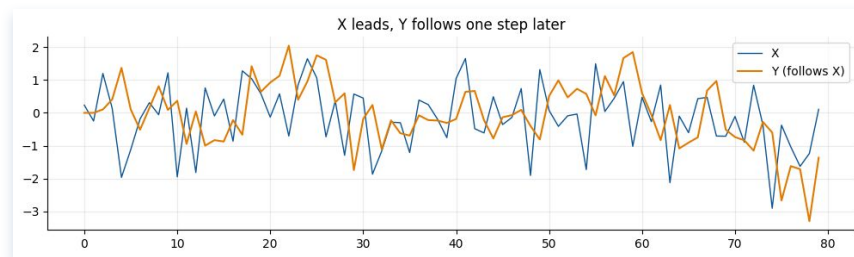
Restricted (Y's past only)

$$Y_t = c + \sum_{i=1}^m \alpha_i Y_{t-i} + \varepsilon_t$$

Unrestricted (add X's past)

$$Y_t = c + \sum_{i=1}^m \alpha_i Y_{t-i} + \sum_{j=1}^m \beta_j X_{t-j} + \varepsilon_t$$

An F-test on the β terms: if adding X's past significantly cuts the prediction error, X Granger-causes Y.

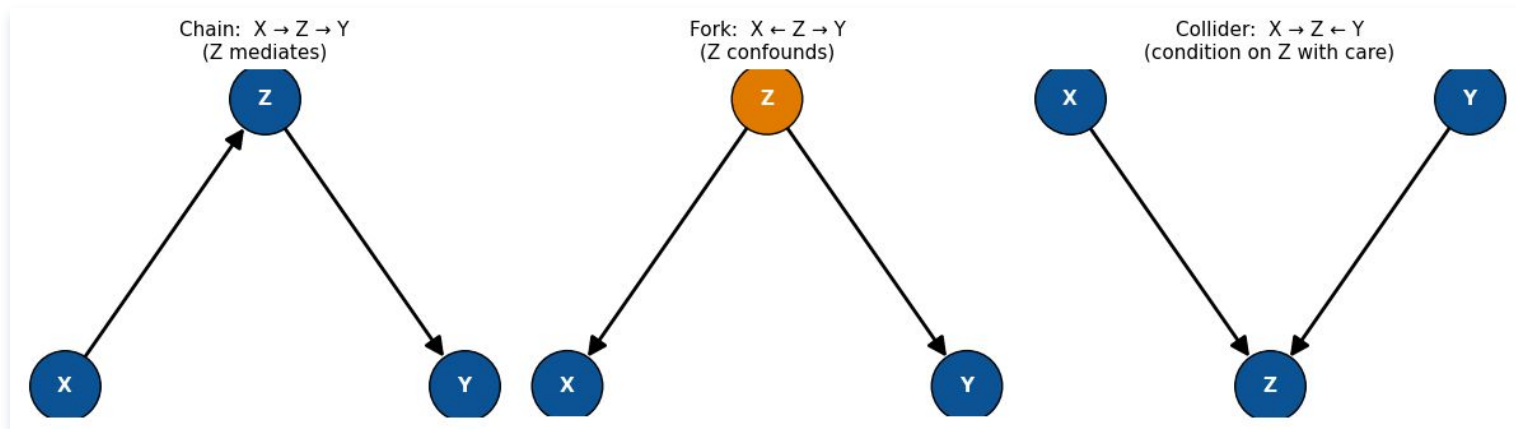


X leads; Y follows one step later — X Granger-causes Y

⚠ Predictive precedence, not proof of true causation — a shared confounder can still fool it.

Causal Directed Acyclic Graphs

Nodes = variables · directed edges = direct causal influence · acyclic = no loops.



Chain

$$X \rightarrow Z \rightarrow Y$$

Z is a mediator: X affects Y through Z.

Fork

$$X \leftarrow Z \rightarrow Y$$

Z is a confounder: creates a spurious X-Y link.

Collider

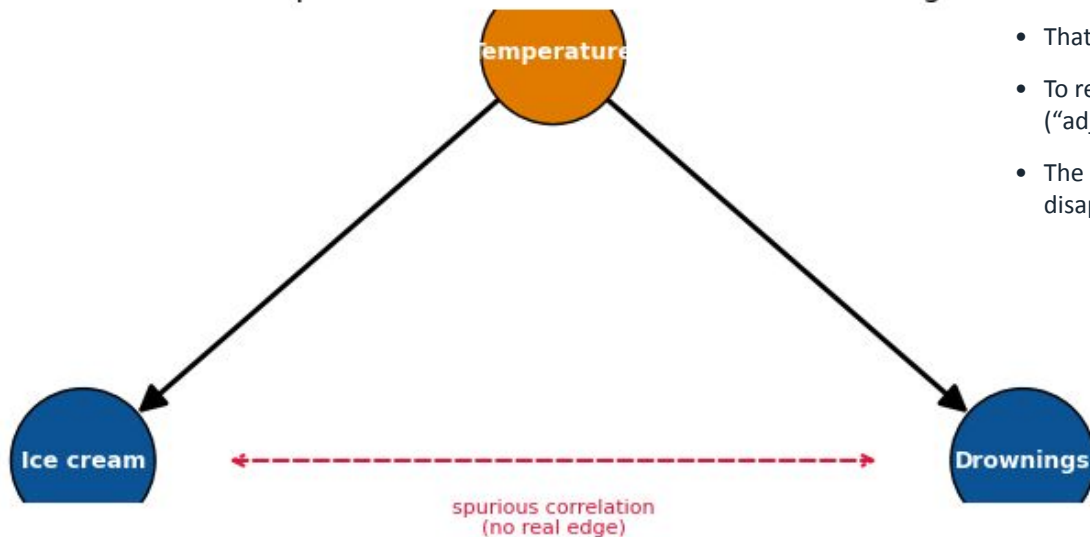
$$X \rightarrow Z \leftarrow Y$$

X, Y independent — until you condition on Z.

Confounders: block the back-door path

The ice-cream example as a fork

Fork: Temperature confounds Ice cream & Drownings



- Temperature $\rightarrow$ ice-cream sales, and Temperature $\rightarrow$ drownings.
- That shared cause is what makes the two outputs correlate.
- To recover the true effect of X on Y, control for the confounder Z ("adjust for temperature").
- The correlation between ice cream and drownings then disappears.

T Time series

- Stationarity = constant mean, variance & lag-only autocovariance. Check with ACF plots and the ADF test.
- ACF/PACF are model fingerprints: AR $\Rightarrow$ PACF cuts off; MA $\Rightarrow$ ACF cuts off.
- ARIMA(p, d, q): p = AR order, d = differences (the “I”), q = MA order.
- AIC / BIC: lower is better; BIC penalises complexity more, favouring simpler models.

C Causality

- Correlation $\neq$ causation — beware reverse causation, confounders and coincidence.
- Granger causality = predictive precedence (past of X helps predict Y); not true causation.
- Causal DAGs encode assumptions as chains, forks (confounders) and colliders.
- To estimate a true effect, block back-door paths by controlling for confounders.

